

Statped Mark-up Requirements

Guidelines for Production of Accessible EPUB3

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1 Introduction

This guidelines document is developed by Statped, a Norwegian governmental agency. Statped Department for Special Education Technology adapts teaching materials for children and adults with special needs, with a focus on students with visual impairments.

These Statped specific guidelines are an addition to the Nordic Guidelines 2015-1. Please refer to the Nordic Guidelines 2015-1 for issues not covered in this document. If there is a conflict between the contents of the Nordic Guidelines 2015-1 and the Statped specific guidelines, please follow the Statped specific guidelines.

For all requirements regarding the EPUB format, please refer to the Nordic Guidelines 2015-1.

A new version of Nordic Guidelines is currently being developed. We will not be using the new Nordic Guidelines yet. For all Statped productions, always follow the Nordic Guidelines 2015-1 until otherwise instructed.

2 Mark-up requirements

These guidelines do not give highly detailed descriptions of how to handle general content. Common recommendations for making valid and accessible HTML content will apply.

As a general rule, the simple solution is almost always the best solution. Using common elements like `<p>`, `<blockquote>`, `<aside>`, `<ul>`, `<ol>`, `<dl>`, `<table>`, `<figure>` etc. and the common structural elements like `<section>` and headings will almost always be sufficient.

More specialised elements or special attributes may be needed occasionally. Some of the more common cases will be described in this document. Less common cases will be covered specifically in editing instructions for the individual productions.

For further information about the common HTML elements and their attributes, please refer to any online resource on HTML 5, for instance: <https://www.w3schools.com>

2.1 Generally

Please do not hesitate to contact us if something is unclear. We appreciate an active dialogue regarding challenging issue.

2.1.1 CSS

Please insert this link to our CSS in the head section of all files:

```
<link rel="stylesheet" type="text/css" href="ebok.css" />
```

2.1.2 Relocation of elements

Certain elements may need to be relocated.

This is to ensure that the flow of the main content is not disrupted.

In cases where more than one type of element needs to be moved, this order should always be followed:

1. **Figures (images)**

2. **Aside elements in general**

`<aside>` elements need to be relocated to ensure that the flow of the main content is not disrupted. Place after any relocated images, before task headings.

Note: Any material that is in some way contained or has a clearly visible beginning and end but is still an integral part of the main text or content, should not be marked up as an `<aside>` element or relocated, but is required to be marked up with `<div>`.

3. **Glossaries/ Description lists (whether placed in an aside or not)**

Place after all other relocated elements.

All relocated elements should be placed before Tasks (where applicable).

2.1.3 Uppercase text

Do not represent headings with uppercase text.

Use an initial capital letter in headings.

In general, do not use capital letters at the start of each word.

In regular text, and text extracted from images, words represented with all uppercase text should be rendered with lower case letters (and with an initial capital letter only at the beginning of a sentence).

In extractions from images (e.g. letter matrix in tasks) single capital letters should be represented in lower case

Exceptions: Proper nouns (must have initial capital letter), abbreviations and acronyms. **Where capital letters represent symbols, or are used in formulas or mathematics expressions, it is very important to preserve the original representation.**

2.1.4 Blank pages in the original source

Please insert the phrase `<p>Blank side.</p>` where the original source has blank pages

2.1.5 Blank pages where elements have been moved

Where elements have been moved to a different page, and left the original page blank, insert the phrase:

`<p>Innhaldet på denne sida har blitt flytta.</p>`
for Norwegian nynorsk language (`xml:lang="nn"`)

`<p>Innholdet på denne siden har blitt flyttet.</p>`
for Norwegian bokmål language (`xml:lang="nb"/"no"`) or other languages

2.1.6 Use of `<em>` and `<strong>`

2.1.6.1 Do not use double emphasis

Never use both `<em>` and `<strong>` to emphasise text, even if text is shown with double emphasis in the original source. If double emphasis occurs, please use only `<strong>`.

2.1.6.2 Headings in `<em>` or `<strong>`

Do not use `<em>` or `<strong>` for headings, even if they appear this way in the printed book.

2.1.6.3 Use of `<em>` or `<strong>` in words and expressions

Where `<em>` or `<strong>` should be used, please make sure that the text is marked up accurately.

Do not include space or punctuation not emphasised in the original.

Punctuation at the end of sentences should not be included in `<em>` or `<strong>` unless the whole sentence is emphasised.

If a sentence ends with an emphasised word, and the next sentence begins with another emphasised word, make sure the words are marked up using separate instances of `<em>` or `<strong>`, and that the end of sentence punctuation is not included.

2.1.6.4 Paragraphs in `<em>` or `<strong>`

As a general rule, do not use `<em>` or `<strong>` for entire paragraphs, even if they appear this way in the printed book.

2.1.6.5 Avoid use of `<em>` or `<strong>` in description lists

Do not use `<em>` or `<strong>` to mark up words within description lists.

2.1.6.6 Avoid use of `<em>` or `<strong>` in table headings

Do not mark up table headings in `<em>` or `<strong>`.

2.1.6.7 Avoid use of `<em>` or `<strong>` in figures and figcaptions

Do not use `<em>` or `<strong>` in figcaptions.

Do not use `<em>` or `<strong>` in text extracted from figures.

2.1.7 Non-breaking space

Always use a non-breaking space between the §-symbol, or other symbols or abbreviations preceding or following numbers (i.e. in currency units or units of measurement). This is important to ensure that the entire expression is kept on the same line, e.g. § 12(1).

Use the code ` ` for non-breaking space.

2.1.8 Table of contents

Delete the images in the toc.

Do not mark up toc as a table.

Do not use all uppercase text in the toc, even if the book uses all uppercase text.

Always place the page reference in the second `<span class="lic">` element.

Example:

```
<ol class="list-style-type-none" style="list-style-type: none;">
  <li><a href="#h1_4"><span class="lic">Kapittel 1: Det
flerkulturelle Norge</span> <span class="lic">9</span></a></li>
```

2.1.8.1 TOC at the beginning of the book

Table of contents should be marked up as shown below unless otherwise instructed:

```
<ol class="list-type-none" style="list-style-type: none;">
```

In cases where the table of contents contains numbering, do not remove the numbering.

Ref. 2015-1 4.18: "Whitespace or other formatting between items in the table of contents should be normalized to **a single space character** between `<span>` elements."

Make sure there is always a single space character between `<span>` elements in the toc.

2.1.8.2 TOC at the beginning of each chapter

If there is a chapter TOC at the beginning of each chapter, please mark up as above:

```
<ol class="list-type-none" style="list-style-type: none;">.
```

2.1.9 Backmatter

2.1.9.1 Indexes and registers

Indexes and registers at the end of a book should be marked up as lists, not as tables.

Lists of image credits at the end of the book should not be included in the file.

2.1.9.2 Answer sections for tasks

Do not include task sections in the backmatter. Backmatter should be limited to the answer section ("Fasit") and any reference sections that follow.

2.1.10 Layout challenges

In those circumstances where text layout creates confusion (for example poem, block quote, list), please ask for clarity.

2.1.11 Use of `<hr>` or `<br>`

Please do not use `<hr>` or `<br>` in mark-up unless specifically instructed to do so. Necessary paragraph breaking or spacing will be addressed in the CSS.

In general avoid the use of `<hr>` to represent purely visual effects.

In lists `<br>` can be used to avoid unnecessary `<p>` elements. See 2.4.3.

2.1.11.1 Use of `<br>` in mathematics books

`<br>` should be used in mathematics books to mark line breaks between text and calculations, or between calculations of expressions.

2.1.12 OCR

Suppliers are required to deliver ninety-nine point ninety-five percent (99.95%) accuracy for text, numeral and symbol content requiring OCR.

Note that accuracy level is measured at the word token level, not per character.

Suppliers are required to deliver one hundred percent (100%) accuracy for text, numeral and symbol content for all other input types where extractable text is available.

Make sure to use the exact same characters as in the original book.

2.2 Thematic grouping of content

To achieve the correct thematic grouping of the content, a careful use of the `<section>` element is important.

2.2.1 Use of section within tasks

When individual tasks have headings, always mark up each task with individual `<section>` elements, and always mark up every task with its own end of section tag.

For further details, see individual editing instructions.

2.3 Figures/Images

2.3.1 Alt tag

Accessibility guidelines require all images to be supplied with a short, descriptive text as value of the `alt` attribute of the `<img>` elements. Suppliers are not required to provide these descriptive texts, but should instead use one of the following generic values:

- `photo` – for photographs
- `illustration` – for illustrations
- `figure` – for schematic images, graphs, diagrams, scientific models etc.
- `symbol` – for icons, signs, inline images etc.
- `map` – for maps
- `drawing` – for drawings
- `comic` – for comic strips and panels
- `logo` – for logos

If there is any doubt about which value to assign to the `alt` attribute, Suppliers are required to use `figure`.

2.3.2 Extraction of text in figures

Text in figures should be included, except in mathematics books.

In general, do not extract text from photos, maps, graphs, images of webpages or pictures of pages from books.

- When only a few words or sentences are extracted, avoid unnecessary use of `<p>`.
- Do not use `<strong>` or `<em>` for text extracted from figures
- Where multiple words or numbers are extracted, please extract text in a systematic and orderly way, e.g. by use of lists (`<ol>/<ul>`) and paragraphs (`<p>`).

2.3.2.1 Extraction of text from figures in mathematics books

In mathematics books, do not extract text from any figures.

Graphs and figures should be represented as images only.

2.3.3 Aside element for image description to be added later

In all figures with a non-empty `alt`-tag, please insert this `<aside>`, with a unique `id`, for descriptions to be added later or for extracted text:

```
<aside class="prodnote" epub:type="z3998:production"
id="desc(x)">⌘</aside>
```

Please note that we have replaced the asterisk `*` used in previous orders with a new symbol `⌘` to avoid confusion with the multiplication sign.

2.3.4 Figcaptions

Do not use `<strong>` or `<em>` in figcaptions.

Avoid unnecessary use of `<p>` in figcaptions.

2.3.5 Placement of figure elements

Figure elements should not disrupt the flow of the main content.

In general, relocate figures before the next heading (after main content, but before other relocated elements e.g. aside elements). More details may be given in editing instructions. Do not relocate figures in mathematics books.

Be aware that in the original book, figures are sometimes placed before the heading and text they belong to.

2.3.5.1 Figures at the start of each chapter, placed before the chapter heading

Sometimes figures at the beginning of each chapter occur on the page preceding the chapter heading. Move these images to the bottom of the page containing the chapter heading, and insert this sentence on the resulting blank page:

```
<p>Kapittelbildet er flyttet til neste side.</p>
```

2.3.6 Figures in science books

Do not relocate figures, only images, in science books.

See individual editing instructions for further details.

2.3.7 Figures in mathematics books

Do not relocate figures and images in mathematics books.

Do not extract text from figures and images. Represent graphs and figures as images only. Please make sure surrounding text is not included in the `<figure>` element.

2.3.7.1 Spreadsheets

Spreadsheets should be marked up as `<table>` and not as `<figure>`.

2.3.8 Comics, comic strips and graphic novels

- Single comic panels should be marked up as `<figure>`.
Please extract the text in an aside, (`aside class="prodnote"` used for extracted text in images.)
Please mark up text in individual paragraphs.
- Comic strips should be marked up as one figure.
Please extract the text in an aside, (`aside class="prodnote"` used for extracted text in images.)
Please mark up text in individual paragraphs.

- Comic strips or graphic novels stretching across one or more pages:
Please mark up each page as one figure, and extract the text in an aside, (`aside class="prodnote"` used for extracted text in images.)
Please mark up text in individual paragraphs.

2.4 Lists

2.4.1 Ordered lists `<ol>`

Please follow the guidelines in 2015-1, 3.1.4.19:

"The `<ol>` tag is used to mark-up ordered lists. Formatting such as numbers or roman numerals that may be associated with the ordered list items is required to be removed unless indicated otherwise by the Ordering Agencies."

The numbering of an ordered list must not be included as content in the `<li>` elements of the list. The default type for the numbering is numeric. This can be changed by using the `type` attribute. [Available list types](#).

The default starting point is 1 (regardless of which type of numbering is used in the ordered list) but can be changed using the `start` attribute. [Available list types](#).

See also use of the `value` attribute under 2.4.1.1

Special requirements may apply for lists in exercises, see 2.5.

2.4.1.1 Lists with missing sequential values

In cases where an ordered list skips a sequential value, please use the `value` attribute on the appropriate `<li>` element(s) to adjust.

Do not divide the `<ol>` into separate lists with `start` attributes.

Example:

```
<ol>
  <li>One</li>
  <li>Two</li>
  <li value="4">Four</li>
  <li>Five</li>
</ol>
```

The `value` attribute should also be used when lists start with any other value than 1 because elements have been excluded (as opposed to a list that simply continues from a previous list).

2.4.1.2 Lists with reversed order

In cases where the order of a list has been reversed, use the `reversed` attribute.

Example:

```
<ol reversed>
  <li>Five</li>
  <li>Four</li>
  <li>Three</li>
  <li>Two</li>
  <li>One</li>
</ol>
```

2.4.1.3 List with non-standard numbering

Mark up lists with non-standard numbering as shown below:

```
<ol class="list-type-none" style="list-style-type: none;">
```

This means that the numbering must be included as content in the `<li>` elements.

Example:



```
<h2>Kapittelinnhold</h2>
  <ol class="list-type-none" style="list-style-type: none;">
    <li>1A Tallregning 8</li>
    <li>1B Regning med potenser 16</li>
    <li>1C Nye potenser 23</li>
    <li>1D Store og små tall - standardform 27</li>
    <li>1E n-terøtter og rotregning 33</li>
  </ol>
```

2.4.1.4 Jointly given answers in lists

In the answer section of mathematics books, answers are sometimes given jointly, as shown in the example below. Please do not mark up answers like 2.10c as a separate `<li>` but include it as text under the task that comes before it.

Example:

2.10

a) 27

b) og c)

Tonnet	Frekvens	Relativ frekvens	Relativ frekvens i prosent
0	8	0,296	29,6 %
1	5	0,185	18,5 %
2	6	0,222	22,2 %
3	2	0,074	7,4 %
4	3	0,111	11,1 %
5	2	0,074	7,4 %
6	1	0,037	3,7 %
Sum	27	0,999	99,9 %

```
<section class="key">
<h2>Oppgave 2.10</h2>
<ol type="a">
  <li>27</li>
  <li>og c) <table> ... </table></li>
</ol>
</section>
```

If more subtasks follow, the `value` attribute should be used to achieve the right numbering.

2.4.2 Unordered lists

Please follow the guidelines in 2015-1, 3.1.4.20:

"The `<ul>` tag is used to mark up unordered lists. Formatting such as bullets or dash characters that may be present in the text node of the list items is required to be removed unless indicated otherwise by the Ordering Agencies."

This means that the bullet marker or dash must not be included as content in the `<li>` elements of the list.

Lists without any bullet markers are required to have the attribute:

```
<ul class="list-unstyled">.
```

2.4.3 Avoid the use of <p> as children of elements

`<p>` elements as children of `<li>` elements should be avoided if possible.

Always consider if `<br>` could be used, unless the `<li>` element contains other block elements like nested lists. See Nordic Guidelines 2015-1, 4.2 Nested lists.

2.4.4 Description Lists

Description lists, e.g. glossaries and word definition lists, should be placed in an `<aside>`. Use the appropriate heading and level `<h[x]>`.

Use the appropriate `class` attribute (e.g. `class="glossary"` for glossary lists).

Make sure that `<aside>` elements containing description lists are placed on the correct level.

When necessary, use the “end-of-section-element” `</section>` to achieve the correct level.

All paired lists of words, phrases, expressions etc. and corresponding definitions, translations etc. are required to be marked up as `<dl>`.

As a general rule, always add : (colon) and a blank space at the end of each `<dt>` as shown below, if not already present in the printed source.

Example: `<dt>Description: </dt><dd>definition</dd>`
resulting in output like this: “Description: definition.”

Note that language attributes may be required, for example with glossaries.

All description lists without a heading in the printed book, must be given a heading. This heading will normally be given in Editing instructions. If not, please use:

“Glossary” for English language (`xml:lang="en"`)

“Ordforklaringar” for Norwegian nynorsk language (`xml:lang="nn"`)

“Ordforklaringer” for Norwegian bokmål language (`xml:lang="nb"/"no"`) or other languages

This does not apply to phrases which are integral to the main text.

2.4.4.1 Phonetics in description lists

If description lists contain phonetic symbols, always mark up as shown below:

`<dt>Description /dɪ'skrɪpʃn/: </dt><dd>definition</dd>`
resulting in output like this: “Description /dɪ'skrɪpʃn/: definition.”

The phonetic representation must belong to the word it describes.

2.4.4.2 Relocation of description lists

Some texts contain multiple description lists in the margin or at the end of each page. Place these description lists together in a single `<aside>` at the end of the appropriate section (ref 2015-1 3.1.4.30, 3.1.4.31).

Use an appropriate heading and level `<h[x]>`.

Use an appropriate class attribute (e.g. `class="glossary"` for glossary lists).

In general, description lists should be relocated after relocated images, after other `aside` elements, and before tasks/exercises.

If description lists structurally belong to a higher-level section, use the end-of-section tag `</section>` to place the `<aside>` containing the description lists on a higher level.

Use an individual heading to mark up the description list(s) from each page.

Use the word “Page” (for books with main language English) or “Side” (for all other main languages) and the page number as the heading.

Place these individual page headings on the level below the main glossary heading.

Example:

`<hx>Side 12</hx>` in books with Norwegian as main language.

`<hx>Page 12:</hx>` in books with English as main language.

Note! In cases where a glossary list belongs exclusively to a box marked up with an `<aside>` element, place the glossary list inside the `<aside>` element. Do not relocate. Use the appropriate `<h[x]>`.

2.5 Tasks

In general, please mark sections containing tasks as `<section class="task">`

We adapt books for students who are blind or suffer from visual impairment.

The individual tasks must be accessible and understandable for use with a braille display or a screen reader.

Please take care to follow our guidelines 2.5.1 regarding markup of task types.

Take care that tables are not used merely as layout tools.

Use lists instead of tables wherever possible. See 2.10.7

In our editing instructions we will refer to two different categories of tasks:

Type 1: There is no need for specialized mark-up.

Please use ordinary list mark-up, `<ol>` or `<ul>`, in accordance with the printed book.

In other books there are tasks with a more complicated structure.

See the points below for detailed instructions on mark up of various task types.

Type 2: Specialized task mark-up. See 2.5.1 below.

2.5.1 Specialized task mark-up

More details will usually be provided in the editing instructions for each book.

2.5.1.1 Main task headings

If a heading is not already supplied in the printed source, the task group should generally be given a main heading:

"Oppgaver" (for `xml:lang="nb"/"no"` or other languages not listed)

"Oppgåver" (for `xml:lang="nn"`)

"Tasks" (for `xml:lang="en"`).

2.5.1.2 Individual task headings

Task groups that contain multiple tasks with extensive or complex content, should be given individual task headings (making it easier to navigate between tasks).

Insert the word:

"Oppgave" (for `"nb"/"no"` or other languages not listed)

"Oppgåve" (for `"nn"`)

"Task" (for `"en"`)

before the task number in the individual task heading

e.g.: "Oppgave 1" (for `"nb"/"no"`) "Oppgåve 1" (for `"nn"`) or "Task 1" (for `"en"`).

2.5.1.3 Subordinate task headings

In cases where tasks contain multiple subordinate tasks (1a, 1 b etc.) with extensive or complex content, these subordinate tasks also need individual headings.

Insert the word:

"Oppgave" (for `"nb"/"no"` or other languages not listed)

"Oppgåve" (for `"nn"`)

"Task" (for `"en"`)

before the task number and letter

e.g.: "Oppgave 1a" (for `"nb"/"no"`) "Oppgåve 1a" (for `"nn"`) or "Task 1a" (for `"en"`).

2.5.1.4 Use of section-elements in tasks

Tasks with individual headings should always be marked up with individual `<section>` elements and given an extra `class` attribute with the value `"task"`: `<section class="task">`.

Make sure every task is marked up with its own end of section.

If a book contains an answer section, the `<section>` elements of the answer section should be given the additional attribute `class="key"` and not `class="task"`.

2.5.1.5 Symbols for task types

In some books, there are symbols representing different task types.

In such cases, please insert corresponding terms describing task type, in place of images containing symbols describing task type.

The image reference should be preserved in comment form.

2.5.1.6 Match problems

Some tasks are match problems.

These exercises require the reader to match the contents of one list (words, expressions or figures) with the corresponding items in another list.

Match lists are typically presented side-by-side.

Match problems should be marked up as separate lists, not as tables

If the lists do not have titles in the book, please use individual titles for the two lists: "Liste 1" and "Liste 2".

- Use `<p><strong>` for list titles
- Use `<ol>` for numbered lists
- Use `<ul>` for lists with bullets
- Use `<ul class="list-unstyled">` for unnumbered lists without bullets

2.5.1.7 Fill-in-the-blank tasks

There are several types of Fill-in-the-blanks tasks in educational books

- Always use `_` (underscore) for individual missing letters.
Make sure that the number of missing letters is represented accurately.
- Always use `....` (four punctuation marks) for missing words.
- Fill-in-the-blank tasks take several forms and can for example be represented using a square.

Empty squares and other forms of blanks should be replaced with four dots `....`

$$\square + 2 = 7$$

```
<span class="asciimath">'.... + 2 = 7'</span>
```

If boxes and other forms of blanks have been filled in, they should simply be removed, while the content is preserved.

$$3 - \square = 12$$

```
<p><span class="asciimath">'3 * 4 = 12'</span></p>
```

- In the cases described above, never place the four punctuation marks in an individual `<p>` element.

2.5.1.8 Fill in the correct form – words given

In some tasks, sentences are missing the correct form or translation of a word. In the source material, these words may be written either before or after the sentence.

Place the word in brackets after the sentence (whether the word is placed before or after in the source material).

If there are several words in the same sentence, place all the words within the same brackets, after the sentence, separated by a comma.

Examples:

`<li>Elle est .... en français. (flink)</li>`

`<li><p>Suddenly, the bushes .... (rustle).</p>`

`<p>Father wolf dropped to the ground, .... to leap at the predator. (prepare)</p></li>`

2.5.1.9 Tasks which include lines for answers

Tasks which include lines for answers should be marked up only as lists of questions. It is not necessary to mark up the line meant for writing the answer in the workbook.

However, for Fill-in-the-blanks tasks always use four punctuation marks (....) to represent missing words or phrases between words. See 2.5.1.7

2.5.1.10 Crossword puzzles

Crossword puzzles should be marked up as an image.

Use two unstyled, unordered lists `<ul class="list-unstyled">`, for the lists of crossword clues.

If the source material does not have titles for the two lists, give each list a title:

`<p>Down</p>` and `<p>Across</p>` for books with main language "en"

`<p>Vannrett</p>` and `<p>Loddrett</p>` for books with all other languages.

2.5.1.11 Tables with one letter in each cell (Word search)

Please remove the table and mark up each row as one `<p>`.

Use lower case letters only.

Insert a blank space between each letter.

Example:

`<h2>Aufgabe 2: Wortsalat</h2>`

`<p>Nedanfor er det gøymt 15 ulike møblar og ting</p>`

`<p>s o f a g s e s s e l</p>`

`<p>t e e r f p f f s t f</p>`

`etc.`

These tables should also be captured as a figure.

2.5.1.12 Tasks with figures

In some tasks, figures are a necessary part of the task. These figures should not be moved; they need to be placed where they are in the source material.

2.5.1.13 Tasks designed as boardgames

Extract the text from the spaces or boxes on the gameboard, and mark up as lists, type `<ol>` or `<ul>`. Each space or box on the gameboard represents one list item `<li>`. Make sure that the list items are presented in the correct order.

2.5.1.14 Unformatted lists without bullets within tasks `<ul class="list-unstyled">`

Some tasks contain lists of sentences, words or figures without bullet points or numbers.

Use `<ul class="list-unstyled">` not `<p>` for unnumbered lists of words, sentences or figures without bullets (or numbers).

2.5.1.15 Lists of tasks with non-standard numbering

Lists of tasks with non-standard numbering should be marked up as below:

```
<ol class="list-type-none" style="list-style-type: none;">.
```

This means that the numbering in lists of tasks with non-standard numbering must be included as content in the `<li>` elements of the list, as described in 2.4.1.3.

2.5.1.16 Tasks where examples of answers are given

In many tasks there are lists of questions.

Sometimes the first few questions have examples of answers.

Include these answers in brackets after the question, within the same `<li>`.

2.5.1.17 Tasks with ticking boxes

Many tasks have ticking boxes which often contain headings.

Do not use tables for this type of tasks.

Instead, mark up the questions or statements as a list.

Exclude the ticking boxes or table columns from the mark-up.

Use a `<p>` to mark up the words from the ticking box heading.

Separate words with a slash `" / "`.

Place the `<p>` before the list of questions or statements.

Ticking boxes are often True/False tasks with headings `"Ja"/"Nei"` or `"Riktig"/"Feil"`.

Use `<p>Ja/Nei</p>` for boxes with `"Ja"` and `"Nei"`.

Use `<p>Riktig/Feil</p>` for boxes with `"Riktig"` and `"Feil"`.

Example:

⑥ Hva har de?

	Ja	Nei
1. Har Lena en bok? _____		
2. Har Ali en blyant? _____		
3. Har Pedro et viskelær? _____		
4. Har Dani blyant og viskelær? _____		
5. Har Amina en bok? _____		

```
<section class="task">
<hx>Oppgave 6. Hva har de?</hx>
<p>Ja/Nei</p>
<ol>
  <li>Har Lena en bok?</li>
  <li>Har Ali en blyant?</li>
  <li>... </li>
etc.

</ol>
</section>
```

2.5.1.18 Tasks with tables better represented as lists

In some tasks the source book uses a table layout where it is not strictly necessary. Please consider using lists. See 2.10.7

When using lists for table content, it may often be necessary to use a semicolon, followed by a blank space “; ” to separate the words from separate columns.

2.5.1.19 Tasks with text between subtasks

Some tasks contain text between subtasks.

Normally this text is an introduction to the following subtask.

In the example below this is indicated by lack of indentation.

Such text should always be marked up using a single `<div class="extra-text">` and moved to the following subtask.

`<div class="extra-text">` should never be part of the first `<li>` element in a list.
`<div class="extra-text">` should always be placed at the beginning of the relevant `<li>` element.

The ordered list should not be divided. Place the additional text in `<p>` elements.

Example:

E58 (Kapittel 2, 3 og 6)

Parklands formel brukes for å beregne hvor mange milliliter væske en pasient med store brannskader skal ha tilført i løpet av de 24 første timene etter en forbrenning.

Parklands formel er $V = 4 \cdot m \cdot A$.

V : antall milliliter væske

m : antall kilogram pasienten veier

A : den prosentandelen av kroppens overflateareal som er forbrent

En pasient veier 63 kg, og 25 % av kroppens overflateareal er forbrent.

a Hvor mange liter væske skal pasienten ha tilført i løpet av de 24 første timene etter forbrenningen?

En annen pasient veier 85 kg. En lege beregner at pasienten skal ha tilført 10 L væske de første 24 timene etter en forbrenning.

b Hvor stor prosentandel av kroppens overflateareal er forbrent hos denne pasienten?

(Eksamen 1P våren 2019)

```
<section class="task">
<h3>E58 (Kapittel 2, 3 og 6)</h3>
  <p>Parklands formel brukes ... <br />
  Parklands formel er ... <br />
  <em>V</em>: antall millimeter væske<br />
  <em>m</em>: antall kilogram pasienten veier<br />
  <em>A</em>: den prosentandelen ... <br />
  En pasient veier ... </p>
  <ol type="a">
    <li>Hvor mange liter ... </li>
    <li><div class="extra-text"><p>En annen pasient ... </p></div>
      <p>Hvor stor prosentandel ... </p></li>
  </ol>
  <p>(Eksamen 1P våren 2019)</p>
</section>
```

2.5.2 Tasks in mathematics books

Always mark up individual tasks with individual `<section>` elements with an individual heading. Use a `class` attribute with the value `task`: `<section class="task">`.

Make sure every task is marked up with its own end of section.

In the answers part, `<section>` elements should be given the attribute `class="key"` and not `class="task"`.

Some task headings may have a capital letter next to the number.

Mark up as shown in this example: `<h3>1.1 U</h3>`.

2.6 Sidebars, text boxes etc.

The `<aside>` element is required to be used for any material that is placed in the margin, breaks the flow of the main text or is in some other way to be considered optional or non-essential.

Ensure that `<aside>` elements are placed/relocated in such a way that the flow of the main content is not disrupted. (If `<aside>` elements need to be relocated, place after any relocated images, before a new `<section>`, glossary or task headings.)

Make sure that `<aside>` boxes and headings are placed on the correct level.

When necessary, use the end-of-section element `</section>` to achieve the correct level.

Any material that is in some way contained or has a clearly visible beginning and end but is still an integral part of the main text or content, is required to be marked up with `<div>`, e.g. examples, definitions, tips, boxes containing words or phrases in exercises etc.

In both cases, the elements are required to have a `class` attribute.

Available `class` attribute values:

- `generisk-ramme`
- `ramme bg-red`
- `ramme bg-blue`
- `ramme bg-yellow`
- `ramme bg-gray`
- `ramme bg-beige`

These `class` attribute values may also be applied to other elements, including `<section>`. Further details will be given in specific editing instructions.

2.6.1 Boxes in mathematics books

The boxes in mathematics books do not normally contain skippable content but are part of the main text. They should therefore be marked up as `<div>` rather than `<aside>` unless otherwise stated in the editing instructions.

Some boxes may require separate `<section>` elements and headings. These will be specified in the editing instructions.

2.7 Comments in the margin

Some texts have comments in the margin.

This may be single words, sentences or paragraphs.

Often there are arrows or lines connecting them to parts of the text.

Use an unordered list `<ul>` to mark up these comments.

Place the list of comments at the end of the text they belong to.

In some cases an alternative markup may be requested in accordance with 2.16.13.

Details will be given in editing instructions.

2.8 Texts with specific styles

2.8.1 Plays and screenplays

Plays and screenplays are normally presented in a specific style.

Follow general requirements and do not retain uppercase or emphasis.

- Insert the value `play` for the `class` attribute on the appropriate `<section>` element:
`<section class="play">`.
- Place the speaker's name in `<span>` elements with the `class` attribute value `speaker`:
`<span class="speaker">`.

Example:

```
<p><span class="speaker">Hilde: </span>Jeg liker deg ikke.</p>
<p><span class="speaker">Hans: </span>Jeg liker ikke deg
heller.</p>
```

- For stage directions use the `class` attribute value `directions`:
`<p class="directions">`.

2.9 Page breaks

Page breaks occurring in the source copy should be represented with one of the following elements:

Inline: `<span epub:type="pagebreak"/>`

Other: `<div epub:type="pagebreak"/>`

Use `<div epub:type="pagebreak"/>` where possible.

2.9.1 Relocation of page breaks

- **In general:** Please make sure that inline page breaks are relocated to the nearest full stop so that page breaks do not break up a sentence.
- **Placement of page break elements at the start of a new section:** Where page breaks occur at the beginning of a new `<section>`, please place the page break element between the start of the new `<section>` and the `<section>` heading, not at the end of the previous `<section>`.
- **Page breaks within lists:** When a page break occurs within a list, always place page break at the end of the last list item on the preceding page. Do not place a page break at the beginning of the first list item on the new page.

2.10 Tables

Tables should in general be marked up according to HTML5 standards.

Use `<thead>` and `<tbody>`.

2.10.1 Table titles

Table titles should be placed in a `<caption>`.

The `<caption>` tag must be inserted immediately after the `<table>` tag.

If the table contains a first cell spanning all columns, the text contained within should be marked up as `<caption>`.

Do not use `colspan` in this instance.

Example:

Oslo-Gardermoen				
Antall personer	1	2	3	4
Beløp å betale per person (kroner)			260	

```
<table>
<caption>Oslo-Gardermoen</caption>
<tbody>
<tr>
  <th>Antall personer</th>
  <td>1</td>
  <td>2</td>
  <td>3</td>
  <td>4</td>
</tr>
<tr>
  <th>Beløp å betale per person (kroner)</th>
  <td></td>
```

```

        <td></td>
        <td>260</td>
        <td></td>
    </tr>
</tbody>
</table>

```

If the table contains an additional "flag cell" in the top, left corner, the text contained within should be marked up as `<caption>`.

Example:

Røyking					
Årstall	2008	2009	2010	2011	2012
Kvinner	17	15	14	13	8

```

<table>
<caption>Røyking</caption>
<tbody>
<tr>
    <th>Årstall</th>
    <td>2008</td>
    <td>2009</td>
    <td>2010</td>
    <td>2011</td>
    <td>2012</td>
</tr>
<tr>
    <th>Kvinner</th>
    <td>17</td>
    <td>15</td>
    <td>14</td>
    <td>13</td>
    <td>8</td>
</tr>
</tbody>
</table>

```

2.10.2 Tables without clear boundaries between rows or columns

Please note that in some tables, the printed book does not display clear boundaries between individual rows or columns.

Take care that cell content is correctly placed in a clearly defined cell.

Avoid placing content that does not belong together (e.g. long lists of words or numbers) within one cell in tables.

2.10.3 Avoid use of or in <th>

Never mark up <th> elements with or .

2.10.4 Avoid use of <p> within table cells

Do not use <p> within table cells unless necessary to structure text.

2.10.5 Lists within tables

Take care that lists within tables are marked up according to 2.4

2.10.6 Tables where all table cells are empty

Mark up column headings and/or row headings as list(s) and remove all the cells. These tables should also be captured as <figure>.

2.10.7 Do not use tables purely as a formatting tool

In some books, especially in sections of text dealing with grammar, tables are regularly used as a formatting tool. The table format is often unnecessary for our users, and the content may be better or more simply represented as a list or several lists.

When using lists for table content, it may often be necessary to use a semicolon followed by a blank space “; ” to separate the words from separate columns.

2.10.8 Spreadsheets as tables

Use <table> with the <class> attribute value spreadsheet to mark up spreadsheets. Do not mark up as <figure>.

Example:

	A	B
1	Verdi nå	kr 4 400 000,00
2	Endring i prosent	6
3	Vekstfaktor	1,06

```
<table class="spreadsheet">
<thead>
<tr>
  <th></th>
  <th>A</th>
  <th>B</th>
</tr>
</thead>
<tbody>
<tr>
  <th>1</th>
  <td>Verdi nå</td>
```

```

        <td><span class="asciimath">' " kr " 4400000'</span></td>
</tr>
<tr>
    <th>2</th>
    <td>Endring i prosent</td>
    <td>6</td>
</tr>
<tr>
    <th>3</th>
    <td>Vekstfaktor</td>
    <td><span class="asciimath">'1,06'</span></td>
</tr>
</tbody>
</table>

```

2.11 Quotations, blockquotes, sources

Sources credited must be placed within the same element as the material the source refers to. Do not place sources in the following element.

This applies to sources after quotations, in image captions, tables etc.

2.12 Footnotes and endnotes

Identifying notes and placing them at logical breaks in the text minimizes their impact on the reading of the primary text.

2.12.1 Footnotes

Place footnotes at the end of the relevant `<section>`.

2.12.2 Endnotes and chapter notes

Endnotes or chapter notes must be placed in a `<section>` element at the end of the content file, before the closing of the top-level `<section>` element.

2.13 Grammar

2.13.1 Sentence analysis

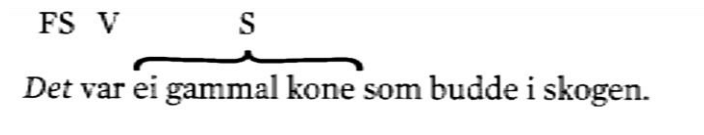
Sentence analysis in grammar often makes use of visual symbols such as brackets and arrows to define and name parts of the sentence correctly, according to syntactic rules.

Sentences with this type of content must be marked up according to a predefined standard.

As a main rule, when no other instructions are given, please represent sentence analysis in text form.

Where words or parts of the sentence are defined as grammatical entities, using brackets or arrows, the sentence must be edited to include the grammatical terms.

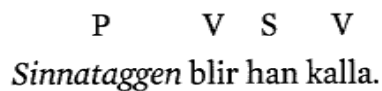
Example 1:



Needs to be represented like this:

<p>[FS: Det] [V: var] [S: ei gammal kone] som budde i skogen.</p>

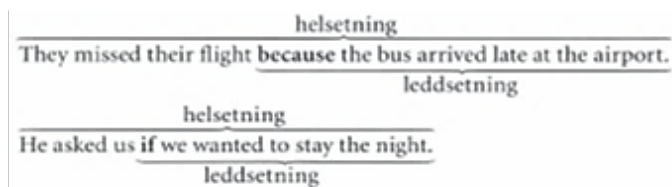
Example 2:



Needs to be represented like this:

<p>[P: Sinnataggen] [V: blir] [S: han] [V: kalla].</p>

Example 3:



<p>They missed their flight because the bus arrived late at the airport.
[helsetning]</p>

<p>because the bus arrived late at the airport. [leddsetning]</p>

<p>He asked us if we wanted to spend the night. [helsetning]</p>

<p>if we wanted to spend the night. [leddsetning]</p>

As an exception, and only if specifically requested by the ordering agency, sentences containing grammatical analysis may be represented as an image with description. In such cases, extracted text must be presented in a logical form, as shown in above.

2.13.2 Conjugation tables

Consider using lists to mark up contents of conjugation tables.

Avoid unnecessary use of the table format. See also 2.10.7

Please also note that in many grammar tables, the printed book does not display clear boundaries between individual rows or columns.

Take care that cell content is correctly placed in a clearly defined cell.

Avoid placing content that does not belong together (e.g. long lists of words or sentences) within a single cell in tables.

2.14 Poems

Follow Nordic Guidelines 2015-1 for markup of poetry.

Example:

```
<section class="poem" epub:type="" id="">
  <hx id=""> ... </hx>
  <div class="linegroup">
    <p class="line"> ... </p>
  </div>
  <p epub:type="z3998:author"> ... </p>
</section>
```

Place author information after the last `linegroup`, before the end of the poem section.

Mark up author information as:

```
<p epub:type="z3998:author"> ... </p>
```

2.15 Inline language markup

Correct inline language markup must be applied throughout the book.

It is especially important that inline language changes are precisely marked up in sections concerning grammar, e.g. sentence analysis, conjugation tables, lists of regular or irregular verbs, translations etc.

2.16 Mathematics

The following guidelines are intended for markup of mathematics books and other books containing mathematics content, e.g. science books.

Always mark up formulas (in chemistry, physics and mathematics) with ASCIIMath.

In science books, excluding mathematics books, do not mark up chemical elements occurring in the regular text with ASCIIMath. Use `<sub>` and `<sup>`, e.g. CO₂:

CO₂.

2.16.1 Decimal numbers and natural numbers

To avoid any issues with decimal separators, always use ASCIIMath for decimal numbers.

Always use ASCIIMath for individual natural numbers from 1000 and higher. Individual natural numbers from 0-999 do not need to be marked up with ASCIIMath.

Task numbering should never be marked up with ASCIIMath.

2.16.2 Time and date

Numbers denoting time or date should not be marked up with ASCIIMath unless they are part of a calculation.

Time spans given in years should not be marked up as ASCIIMath, e.g. 1990–2017.

Typically, time and date are indicated in the following ways in the source material:

- Both time and date are often indicated by natural numbers separated by full stop(s)
Examples: 12.45, 15.20.10 and 2020.07.07.
- A colon is sometimes used to indicate time.
Examples: 12:45 and 15:20:10.
- Strokes are sometimes used to indicate dates.
Example: 2020/7/7.

Calculations with time must be marked up with ASCIIMath.

Use the additional value of `verbatim` for the `class` attribute.

Example:

$$\begin{array}{r} \overset{21}{22}.\overset{60}{22} \\ - 1.47 \\ \hline = 20.35 \end{array}$$

`<span class="asciimath verbatim">'22.22 - 1.47 = 20.35'</span>`

2.16.3 Intervals

Always use ASCIIMath for intervals, like in the example below:

Hvis vi hadde delt intervallet $[150, 160)$ i to deler, $[150, 155)$ og $[155, 160)$, måtte vi fordele de 28 elevene på hver av disse to intervallene. I gjennomsnitt ville det da ha blitt 14 elever i hvert av dem. Vi hadde da fått halve søylehøyden.

`<p>Hvis vi hadde delt intervallet (:150, 160:) i to deler, (:150, 155:) og (:155, 160:), måtte vi fordele de 28 elevene på hver av disse to intervallene.</p>`

2.16.4 Round brackets /parentheses

Do not insert extra, unnecessary round brackets /parentheses not found in the source material.

Example:

$$h(x) = 2x^{1.5}$$

Should be marked up as:

`<span class="asciimath">'h(x) = 2x^1.5'</span>`

Do not use:

`<span class="asciimath">'h(x) = 2x^(1.5)'</span>`

2.16.5 Cancelling

Cancelling, shown by strokes through numbers as in the example below, should not be included. Mark up the numbers only, as if the strokes are not there.

$$\frac{3^5}{3^3} = \frac{3 \cdot 3 \cdot \cancel{3} \cdot \cancel{3} \cdot \cancel{3}}{\cancel{3} \cdot \cancel{3} \cdot \cancel{3}} = \frac{3 \cdot 3}{1} = 3^2 = 9$$

`<span class="asciimath">'3^5/3^3 = (3 * 3 * 3 * 3 * 3)/(3 * 3 * 3) = (3 * 3)/1 = 3^2 = 9'</span>`

2.16.6 Carried and borrowed digits

Carried or borrowed digits should not be included. Mark up the main numbers only, i.e. do not include the carried number "1" in the example below.

$$\begin{array}{r} 1 \\ 14 \\ + 18 \\ \hline = 32 \end{array}$$

`<span class="asciimath">'14 + 18 = 32'</span>`

2.16.7 Fractions

Do not use fraction symbols. For example, $\frac{1}{2}$ should not be converted into $\frac{1}{2}$.

Mark up fractions with ASCIIMath: `<span class="asciimath">'1/2'</span>`

2.16.8 Tally marks



Tally marks should be represented with lower case l's: I, II, III, IIIL, IIILIL.

The number 5 should be represented with five lower case l's even when the source material uses four with a fifth slashed line across.

For numbers higher than 5 divide the tally marks into groups of five separated with a space: IIILIL IIILIL IIILIL.

2.16.9 Do not split calculations of expressions

As a general rule, calculations of expressions should not be split into more than one `<span>` element.

An ASCIIMath expression should normally not start with the equal sign = if it is a continuation of the calculation of the expression on the line above.

Exceptions may occur when there is descriptive text next to calculations. See 2.16.13.

Example:

$$\begin{aligned} &(5x + 2y) + (4x + 6y) - (2x + 3y) \\ &= 5x + 2y + 4x + 6y - 2x - 3y \\ &= 5x + 4x - 2x + 2y + 6y - 3y = 7x + 5y \end{aligned}$$

`<span class="asciimath">'(5x + 2y) + (4x + 6y) - (2x + 3y) = 5x + 2y + 4x + 6y - 2x - 3y = 5x + 4x - 2x + 2y + 6y - 3y = 7x + 5y'</span>`

Where the equal sign = is used in equations to indicate two expressions that have the same value, and these span two or more lines, separate `<span>` elements must be used.

For an exception to this rule see 2.16.13.

2.16.10 Lists of numbers, words or items without separators in mathematics books

Lists of numbers, words or items not containing separators, like commas, should be marked up with `<div>` elements and given a `class` attribute with the value `list-enumeration-1`.

Each individual listing should be marked up as a `<span>` element. Note that in some instances more than one level of `<span>` elements may be necessary, see examples 3 and 4 below.

Include `class="asciimath"` on the `<span>` element in accordance with the requirements for ASCIIMath markup given elsewhere in this document.

Items not on a straight line should be presented in a left-to-right reading order as far as is possible.

Example 1:

1.1 ▣

Skriv tallene i stigende rekkefølge.

a 4 -1 3 -5 -2

b -2 -2,25 -2,255 1,25 1,255

```
<section class="task">
<h3>1.1 U</h3>
<p>Skriv tallene i stigende rekkefølge.</p>
<ol>
  <li>
    <div class="list-enumeration-1">
      <span>4</span>
      <span class="asciimath">' -1 '</span>
      <span>3</span>
      <span class="asciimath">' -5 '</span>
      <span class="asciimath">' -2 '</span>
    </div>
  </li>
  <li>
    <div class="list-enumeration-1">
      <span class="asciimath">' -2 '</span>
      <span class="asciimath">' -2.25 '</span>
```

```

        <span class="asciimath">' -2.255'</span>
        <span class="asciimath">' 1.25'</span>
        <span class="asciimath">' 1.255'</span>
    </div>
</li>
</ol>
</section>

```

Example 2:

E6 (Kapittel 1)

Sorter tallene i stigende rekkefølge.

$0,046 \cdot 10^{11}$ $\frac{46}{1\,000\,000}$ $46 \cdot 10^{-7}$
 $4\,600\,000$ $4,6 \cdot 10^6$ $0,46 \cdot 10^{-6}$

(Eksamen 2P våren 2016)

```

<section class="task">
<h3>E6 (Kapittel 1)</h3>
<p>Sorter tallene i stigende rekkefølge.</p>
<p>
    <div class="list-enumeration-1">
        <span class="asciimath">'0.046 * 10^11'</span>
        <span class="asciimath">'46/1000000'</span>
        <span class="asciimath">' ... '</span>
        <span class="asciimath">' ... '</span>
        <span class="asciimath">' ... '</span>
        <span class="asciimath">'0.46 * 10^-6'</span>
    </div>
</p>
<p>(Eksamen 2P våren 2016)</p>
</section>

```

Example 3:

1.69 5400 og $5,4 \cdot 10^3$ $0,24 \cdot 10^5$ og $2,4 \cdot 10^4$

```

<section class="key">
<h3>1.69</h3>
<p>
    <div class="list-enumeration-1">
        <span>
            <span class="asciimath">'5400'</span> og
            <span class="asciimath">'5.4*10^3'</span>
        </span>
    </div>
</p>

```

```

        <span>
            <span class="asciimath">'0.24*10^5'</span> og
            <span class="asciimath">'2.4*10^4'</span>
        </span>
    </div>
</p>
</section>

```

Example 4:

1.88 E: $27^{\frac{1}{3}}$ C: $\sqrt[4]{1}$ F: $16^{\frac{1}{2}}$ D: $(\sqrt{2})^2$ H: $\sqrt[6]{6^6}$

```

<section class="key">
<h3>1.88</h3>
<p>
    <div class="list-enumeration-1">
        <span>E:
            <span class="asciimath">'27^(1/3) '</span>
        </span>
        <span>C:
            <span class="asciimath"> ... </span>
        </span>
        <span>F:
            <span class="asciimath"> ... </span>
        </span>
        <span>D:
            <span class="asciimath"> ... </span>
        </span>
        <span>H:
            <span class="asciimath"> ... </span>
        </span>
    </div>
</p>
</section>

```

2.16.11 Markup for individual expressions

Use individual `<span>` elements for individual expressions.

Example:

Tallene 1,4 og 1,5 er positive.

```

<p>Tallene <span class="asciimath">'1.4'</span> og <span
class="asciimath">'1.5'</span> er positive.</p>

```

2.16.12 Descriptive and visual information surrounding numbers

Do not include visual symbols such as curly brackets and descriptive information surrounding numbers.

In the example below the descriptive information is "Grunntall", " "Eksponent" , "Potens", "4 + 3 faktorer", " "4 faktorer" and "3 faktorer". Mark up the main numbers only.

$$\text{Grunntall} \rightarrow 2^4 \leftarrow \text{Eksponent}$$

$\underbrace{\hspace{1.5cm}}$
Potens

Hvis vi skal regne ut $2^4 \cdot 2^3$, får vi

$$2^4 \cdot 2^3 = \underbrace{2 \cdot 2 \cdot 2 \cdot 2}_{4 \text{ faktorer}} \cdot \underbrace{2 \cdot 2 \cdot 2}_{3 \text{ faktorer}} = 2^{4+3} = 2^7$$

`<p><span class="asciimath">'2^4'</span></p>`

`<p>Hvis vi skal regne ut <span class="asciimath">'2^4 * 2^3'</span>, får vi<br />`

`<span class="asciimath">'2^4 * 2^3 = 2 * 2 * 2 * 2 * 2 * 2 * 2 = 2^(4+3) = 2^7'</span></p>`

2.16.13 Descriptive text next to calculations or text

If there is descriptive text next to calculations or text, mark up the descriptive text with a `<span>` element and a `class` attribute with the value `description`.

Place the descriptive text after the end of the ASCIIMath element, word, sentence or any other element it belongs to.

Separate descriptive texts should be placed in individual `<span>` elements.

Use a `class` attribute with the value `content-with-description` to mark up content with descriptive text.

If there is a heading before the content, place the `class` attribute value `content-with-description` on the `<section>` element.

If there is no heading or `<section>` element, wrap the `<span>` element(s) in a `<div>` element with the `class` attribute value `content-with-description`.

Do not use the `<div>` element if the content is contained in a single `<p>` element but instead place the `class` attribute directly on the `<p>` element.

The class `attribute` `content-with-description` can also be used on `<li>` elements where appropriate:

```
<li class="content-with-description">Dette er punkt 1 <span
class="description">Her begynner lista.</span></li>
```

Example 1:

- b) Fellesnevneren for brøkene $\frac{1}{2}$ og $\frac{3}{4}$ er 4. Vi multipliserer med 4 på begge sidene av likhetstegnet for å få bort brøkene. Det er det samme som å gange alle leddene med 4.

$$\frac{1}{2}x + 3 = \frac{3}{4}x - 1$$

$$4 \cdot \frac{1}{2}x + 4 \cdot 3 = 4 \cdot \frac{3}{4}x - 4 \cdot 1$$

$$2x + 12 = 3x - 4$$

$$2x - 3x = -4 - 12$$

$$-x = -16$$

$$\underline{x = 16}$$

◀ Trekk sammen leddene på hver side.

◀ Når $-x = -16$, er $x = 16$.

```
</li>
```

```
<p>Fellesnevneren for brøkene <span class="asciimath">'1/2'</span>
og <span class="asciimath">'3/4'</span> er 4. Vi multipliserer med 4 på
begge sidene av likhetstegnet for å få bort brøkene. Det er det samme som
å gange alle leddene med 4.</p>
```

```
<p class="content-with-description">
```

```
<span class="asciimath">'1/2x + 3 = 3/4x - 1'</span><br />
```

```
<span class="asciimath">'4 * 1/2x + 4 * 3 = 4 * 3/4x - 4 *
1'</span><br />
```

```
<span class="asciimath">'2x + 12 = 3x - 4'</span><br />
```

```
<span class="asciimath">'2x - 3x = -4 - 12'</span><br />
```

```
<span class="description">Trekk sammen leddene på hver
side.</span><br />
```

```
<span class="asciimath">' -x = -16'</span><br />
```

```
<span class="description">Når <span class="asciimath">' -x = -
16'</span> er <span class="asciimath">'x = 16'</span>.</span><br />
```

```
<span class="asciimath">'x = 16'</span>
```

```
</p>
```

```
</li>
```

EKSEMPEL 5

Bruk potensreglene til å regne ut.

$$\text{a} \quad 2^3 \cdot 2 \cdot 2^4$$

$$\text{b} \quad \frac{3^5 \cdot 3^4}{3^2}$$

$$\begin{aligned} \text{a} \quad 2^3 \cdot 2 \cdot 2^4 &= 2^3 \cdot 2^1 \cdot 2^4 \\ &= 2^{3+1+4} \\ &= 2^8 \end{aligned}$$

Husk: 2 kan vi skrive som 2^1 .

Regel ①

$$\begin{aligned} \text{b} \quad \frac{3^5 \cdot 3^4}{3^2} &= \frac{3^{5+4}}{3^2} \\ &= \frac{3^9}{3^2} \\ &= 3^{9-2} \\ &= 3^7 \end{aligned}$$

Regel ①

Regel ②

```
<section class="example">
<h4>Eksempel 5</h4>
<p>Bruk potensreglene til å regne ut.</p>
  <ol type="a">
    <li><span class="asciimath">'2^3 * 2 * 2^4'</span></li>
    <li><span class="asciimath">'(3^5 * 3^4)/3^2'</span></li>
  </ol>
  <section class="solution">
    <ol type="a">
      <li class="content-with-description">
        <span class="asciimath">'2^3 * 2 * 2^4 = 2^3 * 2^1
        * 2^4'</span> <span class="description">Husk: 2 kan vi
        skrive som <span
        class="asciimath">'2^1'</span>.</span><br />
        <span class="asciimath">'= 2^(3 + 1 + 4) '</span>
        <span class="description">Regel <span class="rule-
        number">1</span></span>
        <span class="asciimath">'= 2^8'</span>
      </li>
      <li class="content-with-description">
        <span class="asciimath">'(3^5 * 3^4)/3^2 = 3^(5 +
        4)/3^2'</span> <span class="description">Regel <span
        class="rule-number">1</span></span><br />
        <span class="asciimath">'= 3^9/3^2'</span><br />
        <span class="asciimath">'= 3^(9 - 2)'</span> <span
        class="description">Regel <span class="rule-
        number">2</span></span><br />
        <span class="asciimath">'= 3^7'</span>
      </li>
    </ol>
  </section>
</section>
```


Example 3:

EKSEMPEL PÅ DIKTANALYSE

En sårbar beundring

«Jeg er mørket» er del av diktsamlingen *Brønnen utenfor Nachors stad* fra 1966 av Hans Børli. I diktet kan vi lese hvordan mørket ville være som lysene han beundrer, i stedet for å være seg selv. I denne diktanalysen vil jeg se nærmere på hvordan diktet er bygd opp, og hvilke virkemidler som framhever sårbarheten til mørket.

Tittelen «Jeg er mørket» avslører at stemmen i diktet er mørket. Motivet er mørket, som elsker lysene og ville være levende som flammen, men som i stedet er mørket som får lysene til å skinne: «Jeg ville være / levende som flammen, men / jeg er mørket».

Diktet består av to strofer med fri form. Den første har tre verselinjer og den andre har åtte. Gjentakelsene i diktet skaper et rolig tempo. Et eksempel på dette er bruken av ordet *mørket*: «Jeg er *mørket* / Jeg er *mørket*. *Mørket* / som får de tusen svaiende lys / til å skinne så pinefullt klart». Et annet ord som gjentas, er *elsker*. Gjentakelsene og det langsomme tempoet er med på å understreke disse ordene, slik at leseren forstår at de er viktige.

Allerede i tittelen «Jeg er mørket» ser vi et eksempel på personifisering. Mørket kan snakke og har følelser. Et annet språklig bilde er sammenligningen «Jeg ville være levende som flammen», som står i andre strofe. Dette forsterker følelsen av lengsel og sårbarhet.

Sårbarheten forsterkes også ved hjelp av *kontrast*: Lysene

Overskriften sier noe om temaet i diktet. Les mer om overskrifter på s. 68.

Etter å ha presentert diktet skriver forfatteren at resten av analysen skal vise diktets oppbygning og tematikk. Dette gir analysen en tydelig retning. Les mer om innledning på s. 69.

Ved hjelp av fagbegreper kan du forklare hvordan oppbygningen og formen preger diktet. Les mer på s. 71–72.

I hoveddelen bør du bruke relevante sitater fra diktet. Her understreker sitatet både hvem som er stemmen, og hva som er motivet. Les mer om stemme og motiv på s. 71.

<section>

<h2>Eksempel på diktanalyse</h2>

<section class="content-with-description">

<h3>En sårbar beundring Overskriften sier noe om temaet i diktet. Les mer om overskrifter på s. 68.</h3>

<p>«Jeg er mørket» er del av diktsamlingen *Brønnen utenfor Nachors stad* fra 1966 av Hans Børli. I diktet kan vi lese hvordan mørket ville være som lysene han beundrer, i stedet for å være seg selv. I denne diktanalysen vil jeg se nærmere på hvordan diktet er bygd opp, og hvilke virkemidler som framhever sårbarheten til mørket. Etter å ha presentert diktet skriver forfatteren at resten av analysen skal vise diktets oppbygning og tematikk. Dette gir analysen en tydelig retning. Les mer om innledning på s. 69.</p>

<p>Tittelen «Jeg er mørket» avslører at stemmen i diktet er mørket. Motivet er mørket, som elsker lysene og ville være levende som flammen, men som i stedet er mørket som får lysene til å skinne: «Jeg ville være / levende som flammen, men / jeg er mørket».</p>

<p>Diktet består av to strofer med fri form. Ved hjelp av fagbegreper kan du forklare hvordan oppbygningen og formen preger diktet. Les mer på s. 71–72. Den første har tre verselinjer og den andre har åtte. Gjentakelsene i diktet

skaper et rolig tempo. Et eksempel på dette er bruken av ordet mørket: «Jeg er mørket / Jeg er mørket. Mørket / som får de tusen svaiende lys / til å skinne så pinefullt klart».

I hoveddelen bør du bruke relevante sitater fra diktet. Her understreker sitatet både hvem som er stemmen, og hva som er motivet. Les mer om stemme og motiv på s. 71. Et annet ord som gjentas, er elsker. Gjentakelsene og det langsomme tempoet er med på å understreke disse ordene, slik at leseren forstår at de er viktige.

Allerede i tittelen «Jeg er mørket» ser vi et eksempel på personifikasjon. Mørket kan snakke og har følelser. Et annet språklig bilde er sammenligningen «Jeg ville være levende som flammen», som står i andre strofe. Dette forsterker følelsen av lengsel og sårbarhet.

...

Example 4:

Bruk av versebinding bygger opp under jegets følelse av å være knust.

En metafor for hvordan jeget føler seg.

Sammenlikning er det sentrale virkemiddelet i dette diktet. Plasthinnen over den knuste telefon-skjermen sier mye om hva diktets jeg føler om forholdet til duet.

Den korte verselinja deler verselinja i to, både grafisk og innholdsmessig.

Det lyriske jeget henvender seg til et du.

Metafor. Egenskaper ved å rive skjermbeskytteren av en knust skjerm blir overført til jegets følelser når hun skilles fra duet.

Metafor for jegets følelser. Egenskaper ved den knuste skjermen overføres til jegets følelser.

jeg leser meldinger
gjennom en knust skjerm
glassbitene holdes sammen
av en tynn plasthinne
akkurat som lykken jeg føler
når vi er sammen
den henger fast
i deg
du holder den på plass
og tar den med deg
når du drar
når du river deg løs
og jeg skjærer meg
på alle bitene
hver gang jeg sjekker
om jeg er like hel

Fra Dette skjer ikke (2017) av Ida Lórien Ringdal

```
<section class="poem content-with-description" epub:type="" id="">
  <div class="linegroup">
    <p class="line">jeg leser meldinger</p>
    <p class="line">gjennom en knust skjerm <span class="
"description">Bruk av <strong>versebinding</strong> bygger opp under
```

```

jegets følelse av å være knust.</span> <span class="description">En
metafor for hvordan jeget føler seg.</span></p>
  <p class="line">glassbitene holdes sammen</p>
  <p class="line">av en tynn plasthinne</p>
  <p class="line">akkurat som lykken jeg føler</p>
  <p class="line">når vi er sammen</p>
  <p class="line">den henger fast</p>
  <p class="line">i deg <span class="description">
<strong>Sammenlikning</strong> er det sentrale virkemiddelet i dette
diktet. Plasthinnen over den knuste telefonskjermen sier mye om hva
diktets jeg føler om forholdet til duet.</span> <span class=
"description">Den korte verselinja deler verselinja i to, både grafisk og
innholdsmessig.</span> <span class="description"><strong>Det lyriske
jeget</strong> henvender seg til et du.</span></p>
    <p class="line">du holder den på plass</p>
    <p class="line">og tar den med deg</p>
    <p class="line">når du drar <span class="description">
<strong>Metafor</strong>. Egenskaper ved å rive skjermbeskytteren av en
knust skjerm blir overført til jegets følelser når hun skilles fra
duet.</span></p>
      <p class="line">når du river deg løs</p>
      <p class="line">og jeg skjærer meg <span class="description">
<strong>Metafor</strong> for jegets følelser. Egenskaper ved den knuste
skjermen overføres til jegets følelser.</span></p>
        <p class="line">på alle bitene</p>
        <p class="line">hver gang jeg sjekker</p>
        <p class="line">om jeg er like hel</p>
      </div>
    <p>Fra <em>Dette skjer ikke</em> (2017) av Ida Lórien Ringdal</p>
</section>

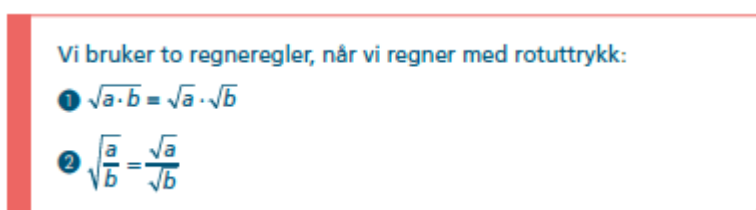
```

2.16.14 Rule numbers

Some books contain rule or reference numbers. Often these reference numbers are represented in dark circles.

Mark up rule or reference numbers with a `<span>` element and a `class` attribute with the value `rule-number`.

Example:



```

<section class="formula">
<p>Vi bruker ... :<br />
<span class="rule-number">1</span> <span class="asciimath">'sqrt(a * ...
'</span><br />
<span class="rule-number">2</span> <span class="asciimath">'sqrt(a/ ...
'</span></p>
</section>

```

Please note that the numbering in the example above is not a list.

2.16.15 Colour coded emphasised numbers

If numbers are emphasised using different colours, please ignore these colours and do not use `<em>` or `<strong>` for emphasis.

2.16.16 Be consistent

If more than one solution is possible, always choose one and apply it consistently throughout the book.